Thresher



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Paddy Thresher; Small-scale seed threshers

What it is used for

A thresher separates grain or seeds from harvested crop stalks, pods, or heads using a rotating threshing mechanism. Use the machine settings and feed rate suitable for the crop to reduce grain damage and maintain safe operation.

Main parts

- Feed chute or crop feeding table
- Threshing drum or rotor with beaters
- Concave, sieve, or separation screen
- Blower and grain collection outlet
- Drive motor or engine and transmission
- Protective guards, belts, and emergency stop

Before you start

1. Confirm that the thresher is suitable for the crop and harvesting condition.
2. Inspect the drum, beaters, screens, belts, guards, and fasteners for damage.
3. Check the drive unit, lubrication, and fuel or electrical supply as applicable.
4. Place the thresher on stable ground and ensure it is properly secured.
5. Clear the work area of stones, metal, ropes, and other foreign objects.
6. Wear close-fitting clothing, eye and hearing protection, suitable footwear, and a dust mask where required.

How to use

1. Keep the power source isolated while inspecting or adjusting the machine.
2. Set the drum speed, concave clearance, and screens according to the crop instructions.
3. Ensure all guards are fitted and the grain collection containers are positioned correctly.
4. Start the machine according to the manufacturer's procedure and allow it to reach normal operating speed.
5. Feed dry or suitably conditioned crop gradually at the recommended rate.
6. Keep hands and tools away from the feed opening and rotating drum.
7. Monitor grain separation, grain damage, chaff discharge, vibration, and unusual noise.
8. Adjust feed rate or approved settings if output becomes uneven or the machine overloads.
9. Stop feeding and allow remaining crop material to clear before shutdown.
10. Switch off the machine and isolate the power before removing blockages or cleaning.
11. Wait until all rotating parts stop completely before opening guards or inspecting the drum.

2. Collect separated grain and remove chaff using safe handling methods.
3. Clean crop residue from the machine and record any faults or maintenance needs.

⚠ Safety

- Only trained operators should operate the thresher.
- Never push crop into the feed opening with hands, feet, or improvised tools.
- Keep guards fitted over belts, pulleys, and rotating components.
- Isolate the power and verify complete stoppage before clearing blockages.
- Keep children and bystanders away from the feeding and discharge areas.
- Control dust and keep sparks, open flames, and smoking away from dry crop residue.

Care & storage

- Remove crop residue from the drum, screens, and discharge paths after use.
- Inspect beaters, screens, belts, bearings, and fasteners regularly.
- Lubricate the specified points and service the drive unit according to instructions.
- Store the machine in a dry, secure location when not in use.
- Arrange repairs for excessive vibration, damaged parts, or poor threshing performance.

Common mistakes to avoid

- Feeding crop too quickly and overloading the drum.
- Using unsuitable drum speed or concave clearance for the crop.
- Clearing a blockage while the machine is running.
- Operating with missing guards or allowing bystanders near the feed opening.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.